

SECTION 040120.64**BRICK – LIMESTONE MASONRY REPOINTING AND STONE CAP REPAIR****PART 1 - GENERAL****1.1 SCOPE OF WORK**

Ste. Genevieve National Historical Park

- A. Project Location: Jean Baptiste Valle - 99 S. Main Street, Sainte Genevieve, MO 63670, Kern House - 19071 US-61, Sainte Genevieve, MO 63670
- B. Building(s) Name/Identifier: Jean Baptiste Valle House & Kern House

1.2 SUMMARY

- A. Section includes reconstructing, repointing and sealing of four historic well caps.
 - 1. Jean Baptiste Valle Well
 - a. Above grade, the well has a limestone cap and is supported by limestone wall. This well should be inspected and any masonry above grade that is lacking mortar or stones, are to be individually replaced and set using an approved mortar mix.
 - 2. Kern House Well #1
 - a. Above grade, this well stands approximately 3 ft tall. The limestone cap is supported by a limestone wall. This well should be inspected and any masonry above grade that is lacking mortar or stones, are to be individually replaced and set using an approved mortar mix.
 - 3. Kern House Well #2
 - a. Above grade, this well stands approximately 2.5 ft tall. The limestone cap is supported by a clay brick wall. The well should be inspected and any masonry above grade that is lacking mortar or brick, is to be individually replaced and set using an approved mortar mix.
 - 4. Kern House Well #3
 - a. Above grade, this well stands approximately 2.5 ft tall. The split limestone cap is supported by a limestone wall. The well should be inspected and any masonry above grade that is lacking mortar or brick, is to be individually replaced and set using an approved mortar mix.
 - 5. Dismantling
 - a. Contractor shall dismantle well by carefully removing the existing cap. Upon removal, masonry structure above grade, shall be inspected for any broken

brick or stones. The contractor shall reuse any existing materials and receive COR approval for a substitute.

- b. Contractors shall only use hand chisels, brushes and other non-powered tools to remove and apply mortar material.
- c. Once the stone/brick walls are reconstructed, the contractor is to set the limestone cap in place with a layer of mortar between the cap and the stone/brick wall.

1.3 PREINSTALLATION MEETINGS

- A. Pre-installation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For each exposed product and for each color and texture specified.
- C. Approved Mortar Mix Proportion Specification.

1.5 INFORMATIONAL SUBMITTALS

- A. Quality-control program.

1.6 QUALITY ASSURANCE

- A. Brick Masonry Repointing Specialist Qualifications: Engage an experienced brick masonry repointing firm to perform work of this Section. Firm shall have completed work similar in material, design, and extent to that indicated for this Project with a record of successful in-service performance. Experience in only installing masonry is insufficient experience for masonry repointing work.
- B. Quality-Control Program: Prepare a written quality-control program for this Project to systematically demonstrate the ability of personnel to properly follow methods and use materials and tools without damaging masonry. Include provisions for supervising performance and preventing damage.
- C. Temperature: Maintain wall temperature between 40 and 95 degrees F. To the extent possible, repoint in the shade, away from strong sunlight in order to slow the drying process.
- D. Mockups: Prepare mockups of brick masonry repointing to demonstrate aesthetic effects and to set quality standards for materials and execution.
 - 1. Repointing: Rake out joints in one area, approximately 12 inches (300 mm) high by 24 inches (600 mm) wide, unless otherwise indicated, for each type of repointing required, and repoint one of the areas.
 - 2. Remove mortar with hand tools (chisel and hammer). Use of power tools is not allowed.

PART 2 - PRODUCTS

2.1 MORTAR MATERIALS

- A. Mortar for restoration shall be Type-O and comply with ASTM C270 and ASTM C1714
- B. Hydrated Lime: Natural Hydraulic Lime (NHL) 3.5
- C. Mortar Sand: ASTM C144.
 - 1. Match size, texture, and gradation of existing mortar sand as closely as possible. Blend several sands if necessary to achieve suitable match.
 - 2. Color: Provide natural sand or ground marble, granite, or other sound stone of color necessary to produce required mortar color.
- D. Water: Potable

2.2 MORTAR MIXES

- A. Measurement and Mixing: Measure cementitious materials and sand in a dry condition by volume or equivalent weight. Do not measure by shovel; use known measure. Mix materials in a clean, mechanical batch mixer.
 - 1. Mixing Pointing Mortar: Thoroughly mix cementitious materials and sand together before adding any water. Then mix again, adding only enough water to produce a damp, unworkable mix that retains its form when pressed into a ball. Maintain mortar in this dampened condition for 15 to 30 minutes. Add remaining water in small portions until mortar reaches desired consistency. Use mortar within one hour of final mixing; do not re-temper or use partially hardened material.

PART 3 - EXECUTION

3.1 REPOINTING

- A. Rake out and repoint joints to the following extent:
 - 1. All joints in areas indicated.
- B. Rake out joints as follows, according to procedures demonstrated in approved mockup:
 - 1. Remove mortar from joints to depth of joint width plus 1/8 inch (3 mm) and not less than that required to expose sound, un-weathered mortar. Do not remove unsound mortar more than 2 inches (50 mm) deep; consult COR for direction.
 - 2. Remove mortar from brick and other masonry surfaces within raked-out joints to provide reveals with square backs and to expose masonry for contact with pointing mortar. Brush, vacuum, or flush joints with clean, low-pressure water spray to remove dirt and loose debris.
 - 3. Do not spall edges of brick or other masonry units or widen joints. Replace or patch damaged brick or other masonry units as directed by COR.

- C. Notify COR of unforeseen detrimental conditions including voids in mortar joints, cracks, loose masonry units, rotted wood, rusted metal, and other deteriorated items.
- D. Pointing with Mortar:
 - 1. Rinse joint surfaces with water to remove dust and mortar particles. Time rinsing application so, at time of pointing, joint surfaces are damp but free of standing water. If rinse water dries, dampen joint surfaces before pointing.
 - 2. Apply pointing mortar first to areas where existing mortar was removed to depths greater than surrounding areas. Apply in layers not greater than 3/8 inch (9 mm) until a uniform depth is formed. Fully compact each layer, and allow it to become thumbprint hard before applying next layer.
 - 3. After deep areas have been filled to same depth as remaining joints, point joints by placing mortar in layers not greater than 3/8 inch (9 mm). Fully compact each layer and allow to become thumbprint hard before applying next layer. Where existing masonry units have worn or rounded edges, slightly recess finished mortar surface below face of masonry to avoid widened joint faces. Take care not to spread mortar beyond joint edges onto exposed masonry surfaces or to featheredge the mortar.
 - 4. When mortar is thumbprint hard, tool joints to match original appearance of joints as approved by COR. Remove excess mortar from edge of joint by brushing.
 - 5. Cure mortar by maintaining in thoroughly damp condition for at least 72 consecutive hours. Coordinate mortar curing accordingly to avoid weekend and holiday work, unless approved by COR.
 - 6. Hairline cracking within mortar or mortar separation at edge of a joint is unacceptable. Completely remove such mortar and repoint.
- E. Where repointing work precedes cleaning of existing masonry, allow mortar to harden at least 30 days before beginning cleaning work.

3.2 FINAL CLEANING OF REPOINTED MASONRY

- A. If repointing work is carefully executed, there will be little need for cleaning other than to remove the small amount of mortar from the edge of the joint following tooling.
- B. Remove excess repointing mortar from joint edge with a stiff natural bristle or nylon brush after the mortar has dried, but before it is initially set. Do not employ aggressive cleaning methods that could damage masonry surfaces or the mortar.

END OF SECTION